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COSC 4661
Project 1
10/24/2026

Project 1: Eagle Delivery

Part A: A* search

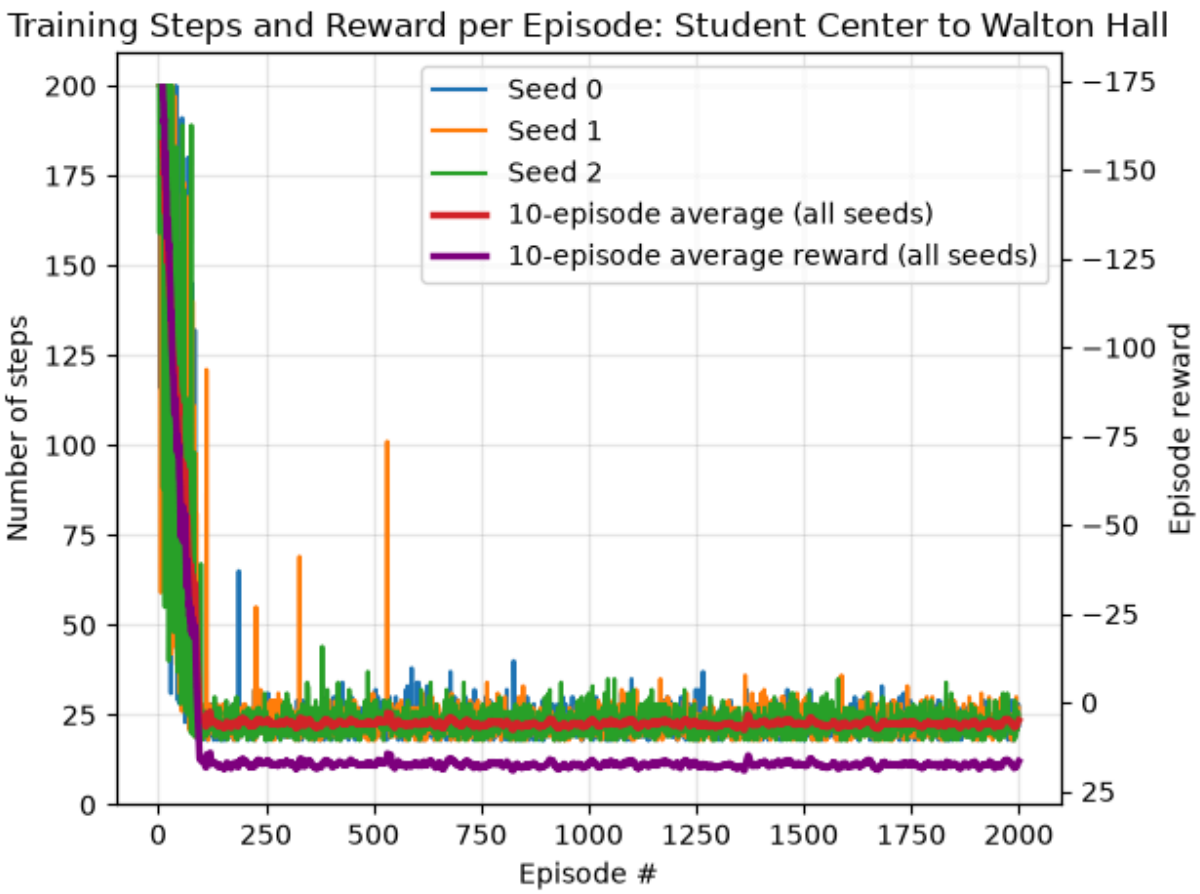
1. Identify state, initial state, actions, transition model, goal test, path cost. Explain why BFS is optimal for the unit cost maps.
   State: agent location(x,y)
   Initial state: location(1,1)
   Actions: Up, Down, Left, Right
   Transition Model: update location
   Goal test: arrived at goal location
   Path cost: -1
2. h(s) = |rs − rg| + |cs − cg| is admissible as a heuristic because every step can only take the courier one step N-S or E-W closer to the goal. This will never overestimate the cost to reach the goal.

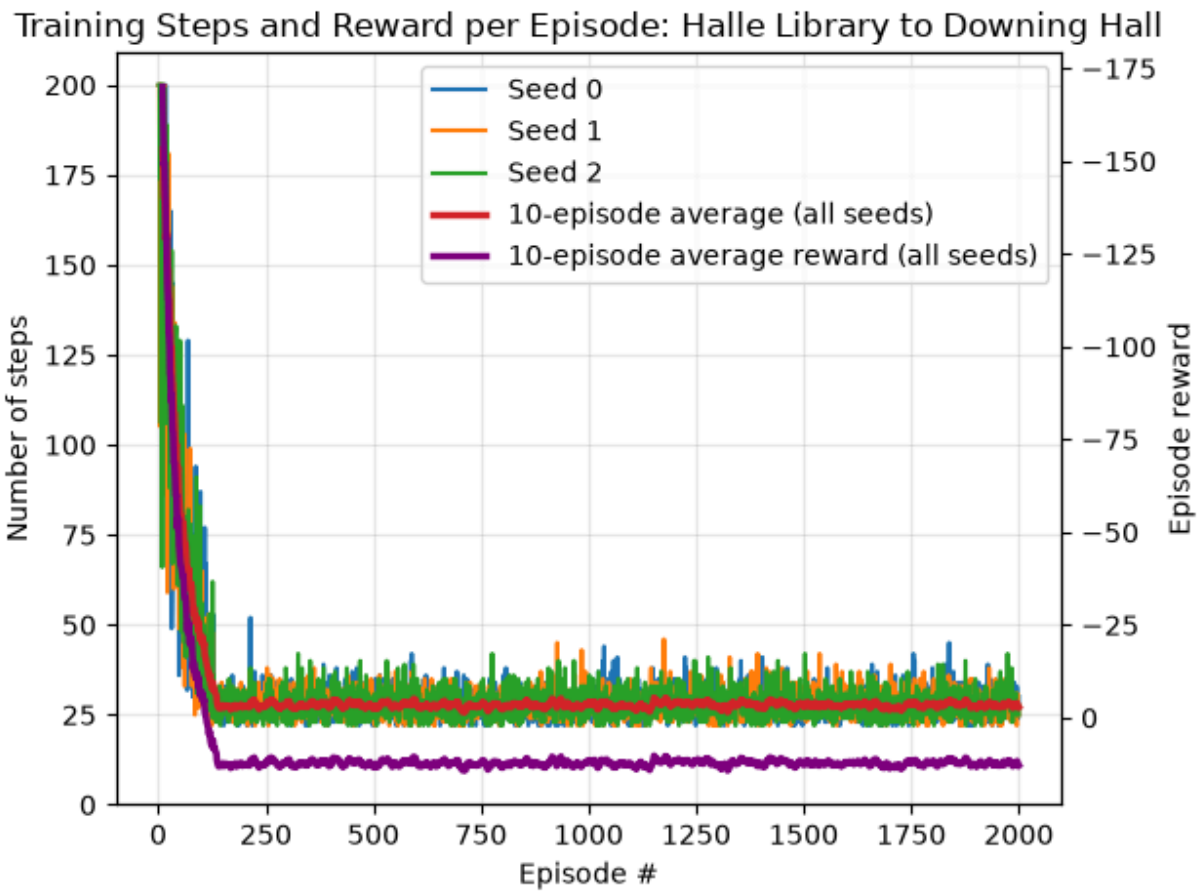
Map	Delivery Task	BFS Route Length	A* Route Length	BFS Expanded Node Count	A* Expanded Node Count
Baseline SW	SW	18	18	67	18
Baseline LD	LD	22	22	79	24
Baseline HP	HP	16	16	41	16
Detour SW	SW	26	26	70	32
Detour LD	LD	22	22	65	24
Detour HP	HP	16	16	41	16

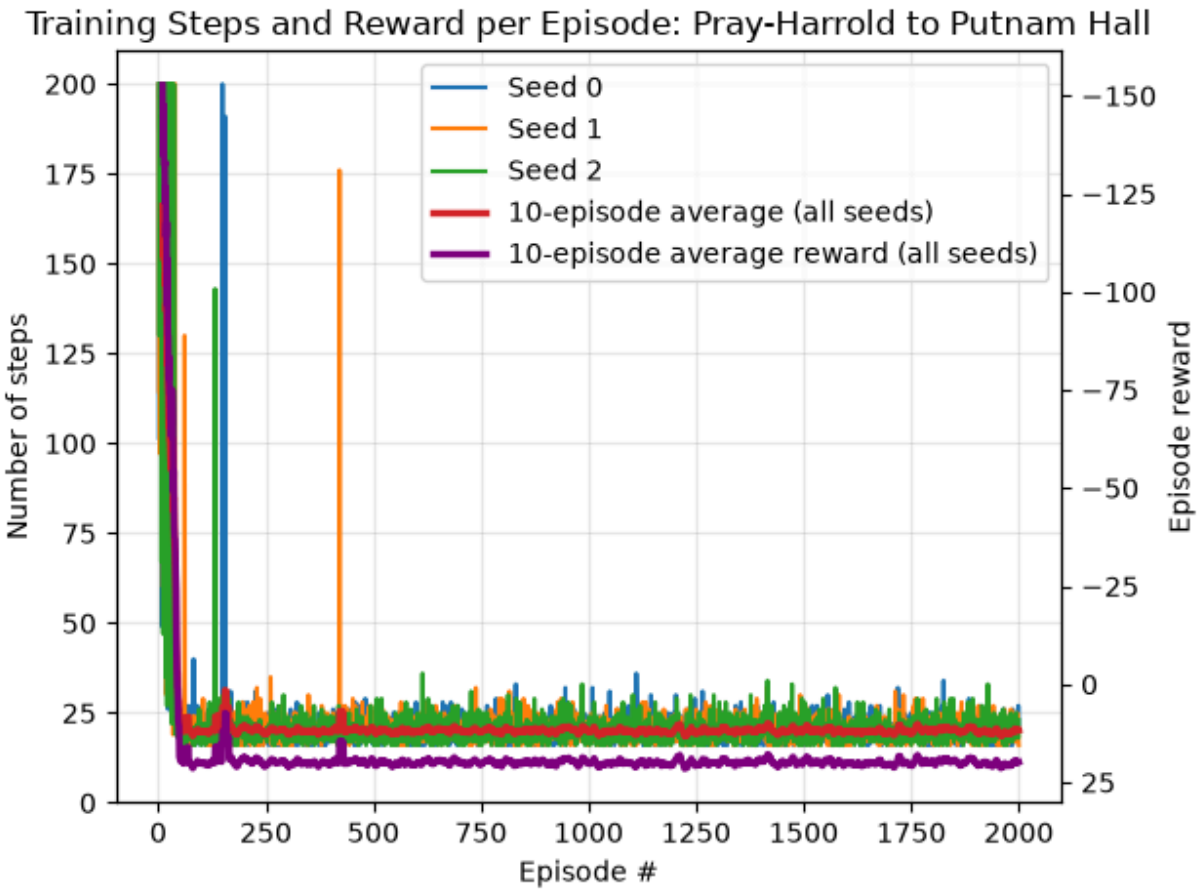
The two algorithms do find paths of equal length, but A* generally expands far fewer nodes.

Part B: Learn a delivery policy

Plots of the delivery training:


Training Steps and Reward per Episode: Student Center to Walton Hall
Seed 0
Seed 1
Seed 2
10-episode average (all seeds)
10-episode average reward (all seeds)
Number of steps
Episode reward
Episode #
0
25
50
75
100
125
150
175
200
−175
−150
−125
−100
−75
−50
−25
0
25
0
250
500
750
1000
1250
1500
1750
2000


Training Steps and Reward per Episode: Halle Library to Downing Hall
Seed 0
Seed 1
Seed 2
10-episode average (all seeds)
10-episode average reward (all seeds)
Number of steps
Episode reward
Episode #
0
25
50
75
100
125
150
175
200
−175
−150
−125
−100
−75
−50
−25
0
0
250
500
750
1000
1250
1500
1750
2000



The training algorithm was able to find a reliable path to the destination within about 150 steps for all three routes, with very little change in the number of steps to the destination afterwards; additional exploration did not decrease the average solution. This is also reflected in the average rewards. After about 150 episodes, the rewards reach a fairly consistent maximum (rewards y-axis is inverted).

Part C: Design your own map

Map	Delivery Task	BFS Route Length	A* Route Length	BFS Expanded Node Count	A* Expanded Node Count
my_campus	SW	21	25	81	74

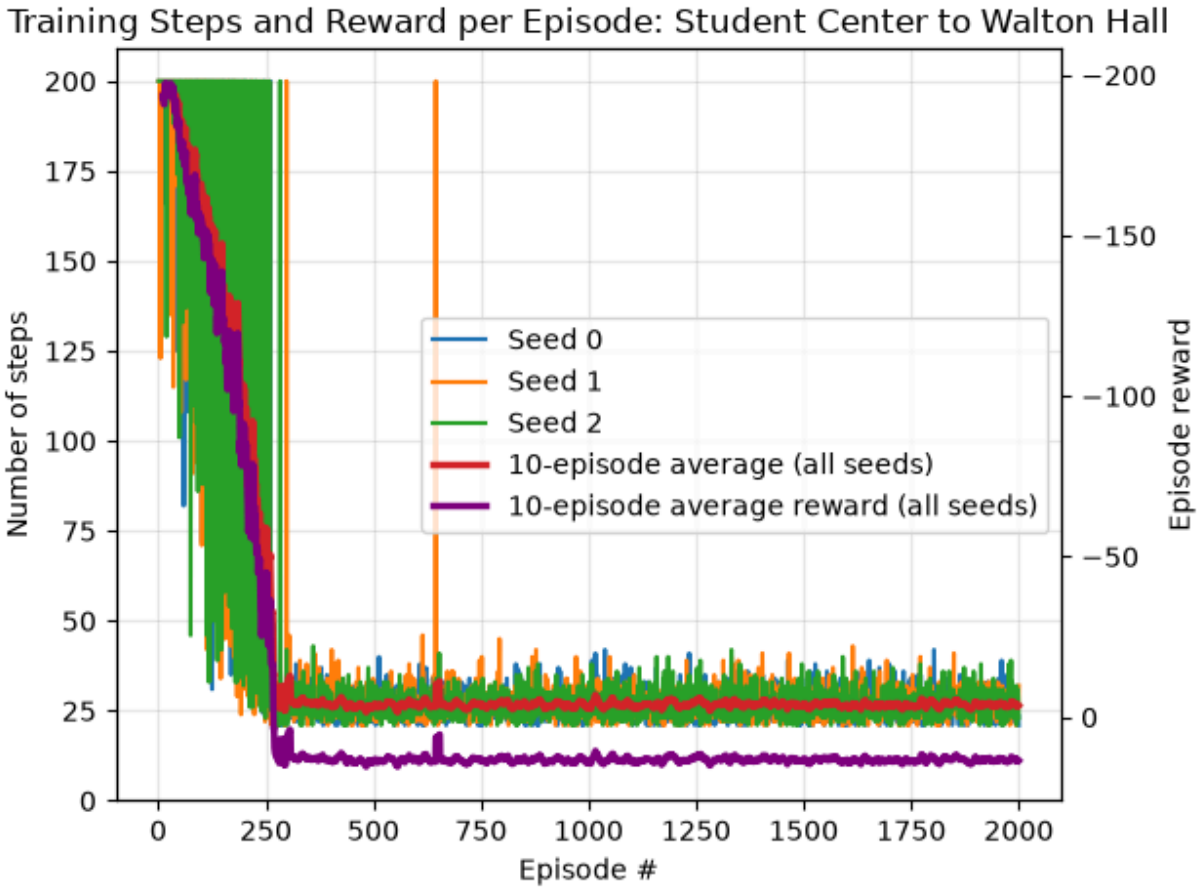
Explain your experiment:

I am going to test changing the alpha value. Lower values of alpha should correspond to slower training rate but more consistent results. Higher alpha values should make the model train faster, but be more unstable in its training results.

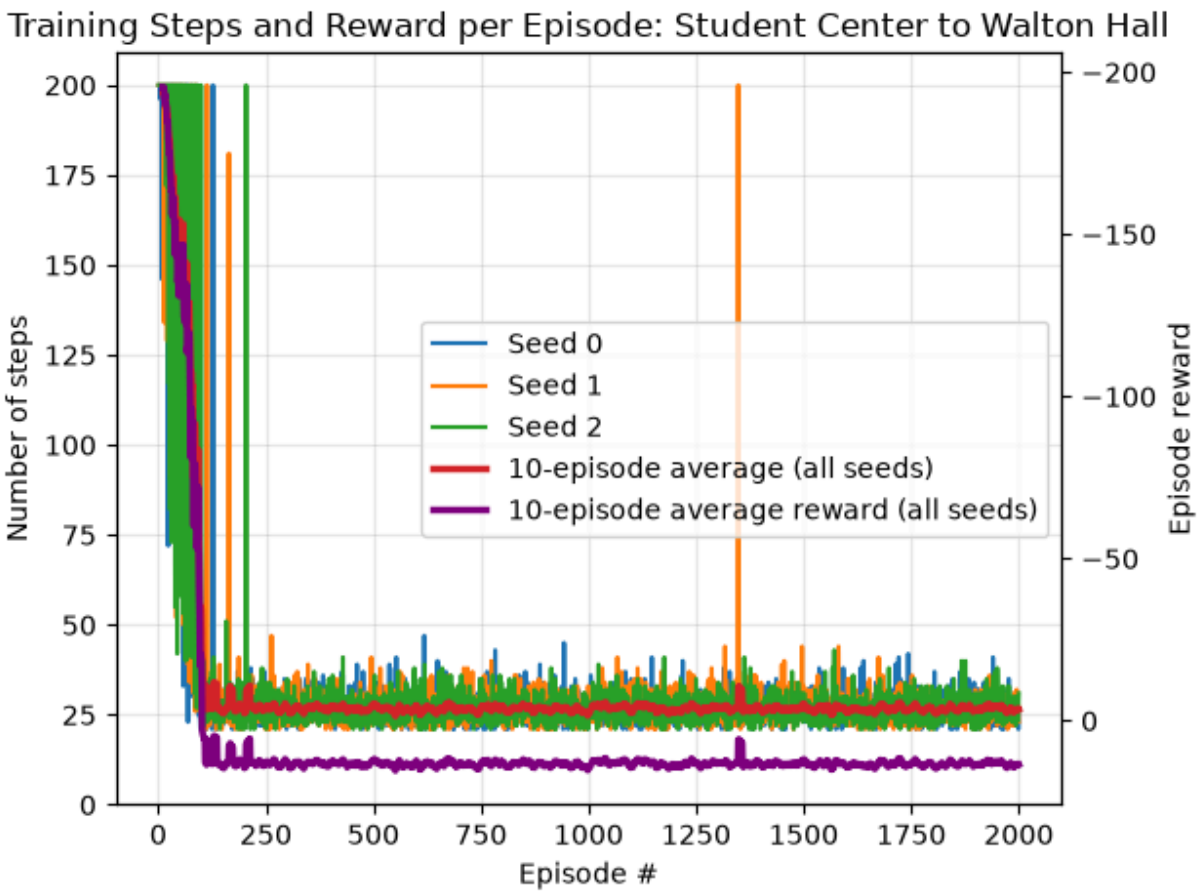
My custom map has S and W on opposite corners. The ideal route heads almost due south before moving east in row 9. The model should discover and learn to ignore the loop in the middle and the dead end on the right side of the map. Interestingly, BFS found the more efficient way of moving south compared to A*.

In my experiment, the only value I changed was alpha: set to be lower than default at 0.1 and higher than default at 0.5. I left all other training parameters the same.

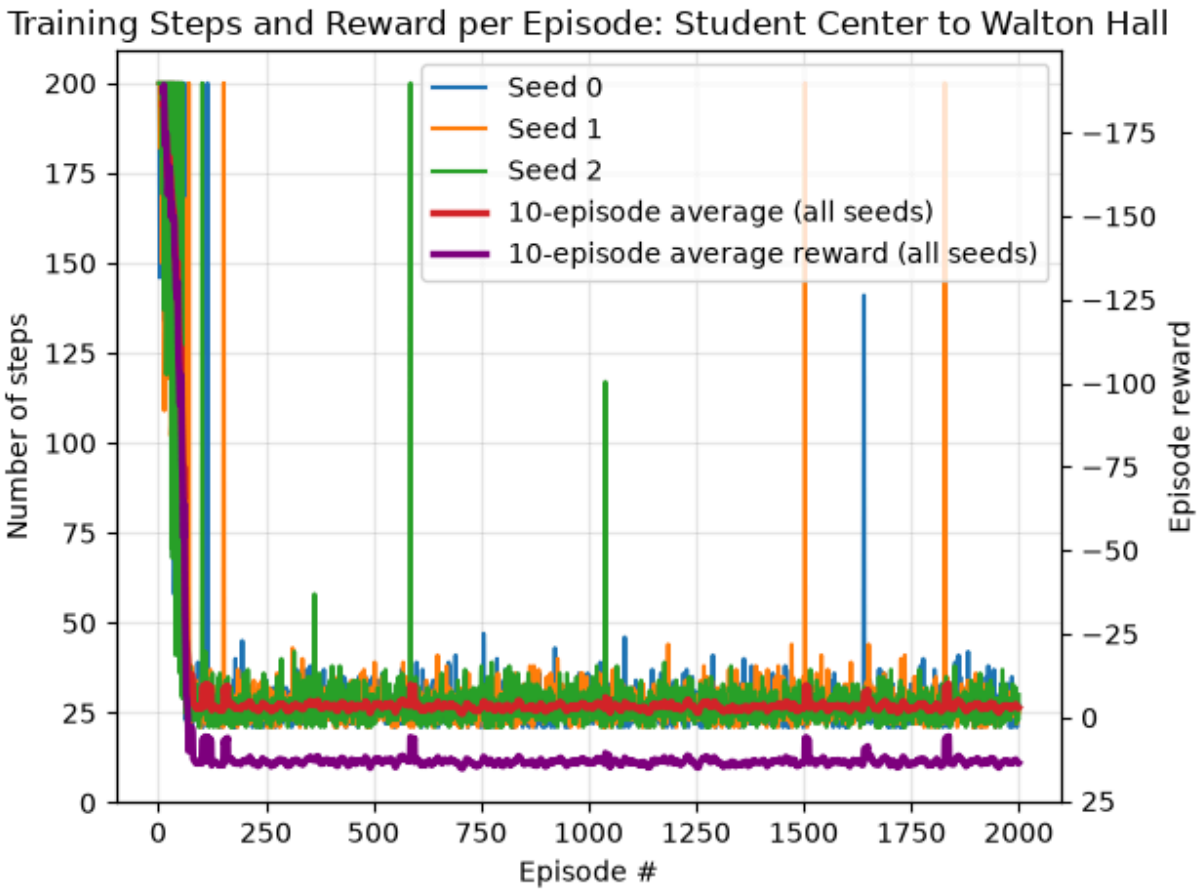
Alpha 0.1



Alpha 0.3


Training Steps and Reward per Episode: Student Center to Walton Hall
Number of steps
Episode reward
Episode #
Seed 0
Seed 1
Seed 2
10-episode average (all seeds)
10-episode average reward (all seeds)
0
25
50
75
100
125
150
175
200
−200
−150
−100
−50
0
0
250
500
750
1000
1250
1500
1750
2000

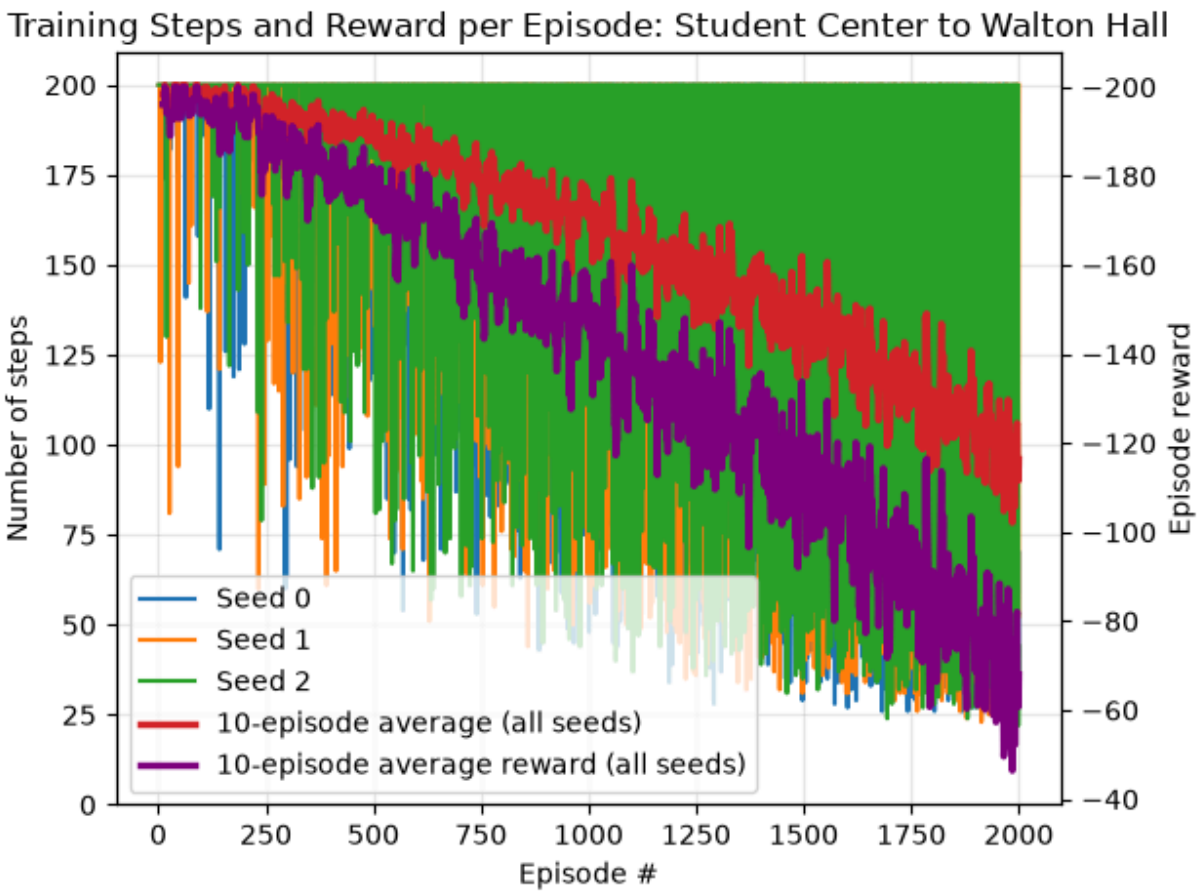
Alpha 0.5



The lower alpha value did decrease training rate, though not dramatically. Once the results had stabilized, they were not noticeably more stable than the default 0.3. For alpha = 0.5, the training rate was faster but not exceptionally so. Once the solution stabilized, the individual seeds also had more step spikes (step counts much higher than the real solution) compared to the lower alpha values.

Out of curiosity, I tried setting alpha to 0.01. This showed the dramatic slowdown in training rate. Even after 2000 episodes, it was not close to reaching the ideal solution, and the individual episode solutions are all over the place.

Alpha = 0.01


Training Steps and Reward per Episode: Student Center to Walton Hall
Number of steps
Episode reward
Episode #
0
25
50
75
100
125
150
175
200
−40
−60
−80
−100
−120
−140
−160
−180
−200
0
250
500
750
1000
1250
1500
1750
2000
Seed 0
Seed 1
Seed 2
10-episode average (all seeds)
10-episode average reward (all seeds)